

Number Theory

The Integer and Division

Theorem 2: THE DIVISION ALGORITHM Let a be an integer and d a positive integer. Then there are unique integers q and r , with $0 \leq r < d$, such that $a = dq + r$

Definition 2: In the equality given in the division algorithm, d is called the *divisor*, a is called the *dividend*, q is called the *quotient*, and r is called the *remainder*. These notations are used to express the quotient and remainder:

$$q = a \text{ div } d, r = a \text{ mod } d.$$

Example: What are the quotient and remainder when 101 is divided by 11?

Solution:

$$101 = 11 \cdot 9 + 2$$

The quotient is $101 \text{ div } 11 = 9$

The remainder is $101 \text{ mod } 11 = 2$

Example: What are the quotient and remainder when -11 is divided by 3?

Solution:

$$-11 = 3(-4) + 1$$

The quotient is $-11 \text{ div } 3 = -4$

The remainder is $-11 \text{ mod } 3 = 1$

Number	Divisibility Rule	Example
2	If the last digit is even (0, 2, 4, 6, 8)	246 → last digit 6 → divisible
3	If the sum of digits is divisible by 3	258 → $2+5+8=15$ → divisible
4	If the last two digits form a number divisible by 4	932 → $32 \div 4 = 8$ → divisible
5	If the last digit is 0 or 5	475 → last digit 5 → divisible
6	If divisible by both 2 and 3	114 → even, digit sum=6 → divisible
7	Double the last digit and subtract from the rest. Repeat if needed	203 → $20 - (2 \times 3) = 14$ → divisible
8	If the last three digits form a number divisible by 8	7352 → $352 \div 8 = 44$ → divisible
9	If the sum of digits is divisible by 9	4905 → $4+9+0+5=18$ → divisible
10	If the last digit is 0	670 → last digit 0 → divisible
11	If the difference between the sum of digits in odd and even positions is divisible by 11	2728 → $(2+2) - (7+8) = 4 - 15 = -11$ → divisible

Modular Arithmetic

Definition 3: If a and b are integers and m is a positive integer, then a is *congruent to b modulo m* if m divides $a - b$. We use the notation $a \equiv b \pmod{m}$ to indicate that a is congruent to b modulo m . If a and b are not congruent modulo m , we write $a \not\equiv b \pmod{m}$.

Theorem 3: Let a and b be integers, and let m be a positive integer. Then $a \equiv b \pmod{m}$ if and only if $a \bmod m = b \bmod m$.

Example: Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6.

Solution:

6 divides $(17 - 5) = 12$. $\therefore 17 \equiv 5 \pmod{6}$

6 does not divide $(24 - 14) = 10$. $\therefore 24 \not\equiv 14 \pmod{6}$

Theorem 4: Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Proof: Omitted.

The set of all integers congruent to an integer a modulo m is called the **congruence class** of a modulo m .

Theorem 5: Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Proof: Omitted.

Example: Because $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$, it follows from Theorem 5 that

$$7 + 11 = 18 \equiv 2 + 1 = 3 \pmod{5}$$

$$7 \cdot 11 = 77 \equiv 2 \cdot 1 = 2 \pmod{5}$$

Corollary 2: Let m be a positive integer and let a and b be integers. Then

$$(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$$

and

$$ab \bmod m = ((a \bmod m)(b \bmod m)) \bmod m.$$

Proof: Omitted.

Example:

$$(8 + 14) \bmod 5 = 22 \bmod 5 = 2$$

$$((8 \bmod 5) + (14 \bmod 5)) \bmod 5 = (3 + 4) \bmod 5 = 7 \bmod 5 = 2$$

$$(8 \cdot 14) \bmod 5 = 112 \bmod 5 = 2$$

$$((8 \bmod 5)(14 \bmod 5)) \bmod 5 = (3 \cdot 4) \bmod 5 = 12 \bmod 5 = 2$$

Application of Congruences

Example: Pseudorandom Numbers

The most commonly used procedure for generating pseudorandom numbers is the **linear congruential method**.

We choose four integers: the **modulus** m , **multiplier** a , **increment** c , and **seed** x_0 , with $2 \leq a < m$, $0 \leq c < m$, and $0 \leq x_0 < m$.

We generate a sequence of pseudorandom numbers $\{x_n\}$, with $0 \leq x_n < m$ for all n , by successively using the congruence

$$x_{n+1} = (ax_n + c) \bmod m.$$

Example: Pseudorandom Numbers (contd.)

Let $m = 9$, $a = 7$, $c = 4$, and $x_0 = 3$

$$x_{n+1} = (ax_n + c) \bmod m$$

$$x_0 = 3$$

$$x_1 = (7x_0 + 4) \bmod 9 = (7 \cdot 3 + 4) \bmod 9 = 25 \bmod 9 = 7$$

$$x_2 = (7x_1 + 4) \bmod 9 = (7 \cdot 7 + 4) \bmod 9 = 53 \bmod 9 = 8$$

$$x_3 = (7x_2 + 4) \bmod 9 = (7 \cdot 8 + 4) \bmod 9 = 60 \bmod 9 = 6$$

$$x_4 = (7x_3 + 4) \bmod 9 = (7 \cdot 6 + 4) \bmod 9 = 46 \bmod 9 = 1$$

$$x_5 = (7x_4 + 4) \bmod 9 = (7 \cdot 1 + 4) \bmod 9 = 11 \bmod 9 = 2$$

$$x_6 = (7x_5 + 4) \bmod 9 = (7 \cdot 2 + 4) \bmod 9 = 18 \bmod 9 = 0$$

$$x_7 = (7x_6 + 4) \bmod 9 = (7 \cdot 0 + 4) \bmod 9 = 4 \bmod 9 = 4$$

$$x_8 = (7x_7 + 4) \bmod 9 = (7 \cdot 4 + 4) \bmod 9 = 32 \bmod 9 = 5$$

$$x_9 = (7x_8 + 4) \bmod 9 = (7 \cdot 5 + 4) \bmod 9 = 39 \bmod 9 = 3$$

$$x_{10} = (7x_9 + 4) \bmod 9 = (7 \cdot 3 + 4) \bmod 9 = 25 \bmod 9 = 7$$

Primes

Definition 1: A positive integer p greater than 1 is called a *prime* if the only positive factors of p are 1 and p . A positive integer that is greater than 1 and is not prime is called *composite*.

Example: The integer 7 is prime since its only positive factors are 1 and 7, whereas the integer 9 is composite since it is divisible by 3.

Theorem 1: THE FUNDAMENTAL THEOREM OF ARITHMETIC Every positive integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of non-decreasing size.

Proof: Omitted.

Example: The prime factorizations of 100, 641, 999, and 1024 are given by

$$100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$$

$$641 = 641$$

$$999 = 3 \cdot 3 \cdot 3 \cdot 37 = 3^3 \cdot 37^1$$

$$1024 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^{10}$$

Theorem 2: If n is a composite integer, then n has at least one prime divisor less than or equal to $\sqrt{n}$.

Proof. Omitted.

Example: Show that 101 is prime.

Solution:

The only primes not exceeding $\sqrt{101} = 10$ are 2, 3, 5, and 7. Since 101 is not divisible by 2, 3, 5, or 7, it follows that 101 is prime.

Example: Find the prime factorization of 7007.

Solution:

First, perform divisions of 7007 by successive primes, beginning with 2 but not exceeding $\sqrt{7007} = 83$:

$$7007 / 7 = 1001$$

Next, divide 1001 by successive primes, beginning with 7 but not exceeding $\sqrt{1001} = 31$:

$$1001 / 7 = 143$$

Next, divide 143 by successive primes, beginning with 7 but not exceeding $\sqrt{143} = 11$:

$$143 / 11 = 13$$

Next, divide 13 by successive primes, beginning with 11 but not exceeding $\sqrt{13} = 3$, which is not possible. The procedure is completed.

$$\text{Therefore, } 7007 = 7 \cdot 7 \cdot 11 \cdot 13 = 7^2 \cdot 11^1 \cdot 13^1$$

Greatest Common Divisors and Least Common Multiples

Definition 2: Let a and b be integers, not both zero. The largest integer d such that $d \mid a$ and $d \mid b$ is called the *greatest common divisor* of a and b . The greatest common divisor of a and b is denoted by $\gcd(a, b)$.

Example: What is the greatest common divisor of 24 and 36?

Solution:

The positive common divisors of 24 and 36 are 1, 2, 3, 4, 6, and 12. Hence, $\gcd(24, 36) = 12$.

Example: What is the greatest common divisor of 17 and 22?

Solution:

The only common divisor of 17 and 22 is 1, hence $\gcd(17, 22) = 1$.

Definition 3: The integers a and b are *relatively prime* if their greatest common divisor is 1.

Example: The integers 17 and 22 are relatively prime, since $\gcd(17, 22) = 1$.

Definition 4: The integers $a_1, a_2, \dots, a_n$ are *pair wise relatively prime* if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.

Example: Determine whether the integers 10, 17, and 21 are pairwise relatively prime and whether the integers 10, 19, and 24 are pairwise relatively prime.

Solution:

$\gcd(10, 17) = 1$, $\gcd(10, 21) = 1$, and $\gcd(17, 21) = 1$. Therefore, 10, 17, and 21 are pairwise relatively prime.

$\gcd(10, 24) = 2 > 1$. Therefore, 10, 19, and 24 are not pairwise relatively prime.

One way to find the greatest common divisor of two integers is to use the prime factorizations of these integers.

Suppose that the prime factorizations of the integers a and b , neither equal to zero, are

$$a = p_1^{a_1} p_2^{a_2} p_n^{a_n}$$

$$b = p_1^{b_1} p_2^{b_2} p_n^{b_n}$$

where exponent is a nonnegative integer, and where all primes occurring in the prime factorization of either a or b are included in both factorization, with zero exponents if necessary. Then

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_n^{\min(a_n, b_n)}$$

Example: The prime factorization of 120 and 500 are

$$120 = 2^3 \cdot 3^1 \cdot 5^1$$

$$500 = 2^2 \cdot 3^0 \cdot 5^3$$

$$\gcd(120, 500) = 2^{\min(3, 2)} 3^{\min(1, 0)} 5^{\min(1, 3)} = 2^2 3^0 5^1 = 20$$

Definition 5: The least common multiple of the positive integers a and b is the smallest positive integer that is divisible by both a and b . The least common multiple of a and b is denoted by $\text{lcm}(a, b)$.

Suppose that the prime factorizations of the integers a and b , neither equal to zero, are

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}$$

$$b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n}$$

where exponent is a nonnegative integer, and where all primes occurring in the prime factorization of either a or b are included in both factorization, with zero exponents if necessary. Then

$$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \cdots p_n^{\max(a_n, b_n)}$$

Example: What is the least common multiple of $95256 = 2^3 \cdot 3^5 \cdot 7^2$ and $432 = 2^4 \cdot 3^3$?

Solution:

$$\text{lcm}(2^3 \cdot 3^5 \cdot 7^2, 2^4 \cdot 3^3) = \text{lcm}(2^3 \cdot 3^5 \cdot 7^2, 2^4 \cdot 3^3 \cdot 7^0) = 2^{\max(3, 4)} 3^{\max(5, 3)} 7^{\max(2, 0)} = 2^4 \cdot 3^5 \cdot 7^2 = 190512$$